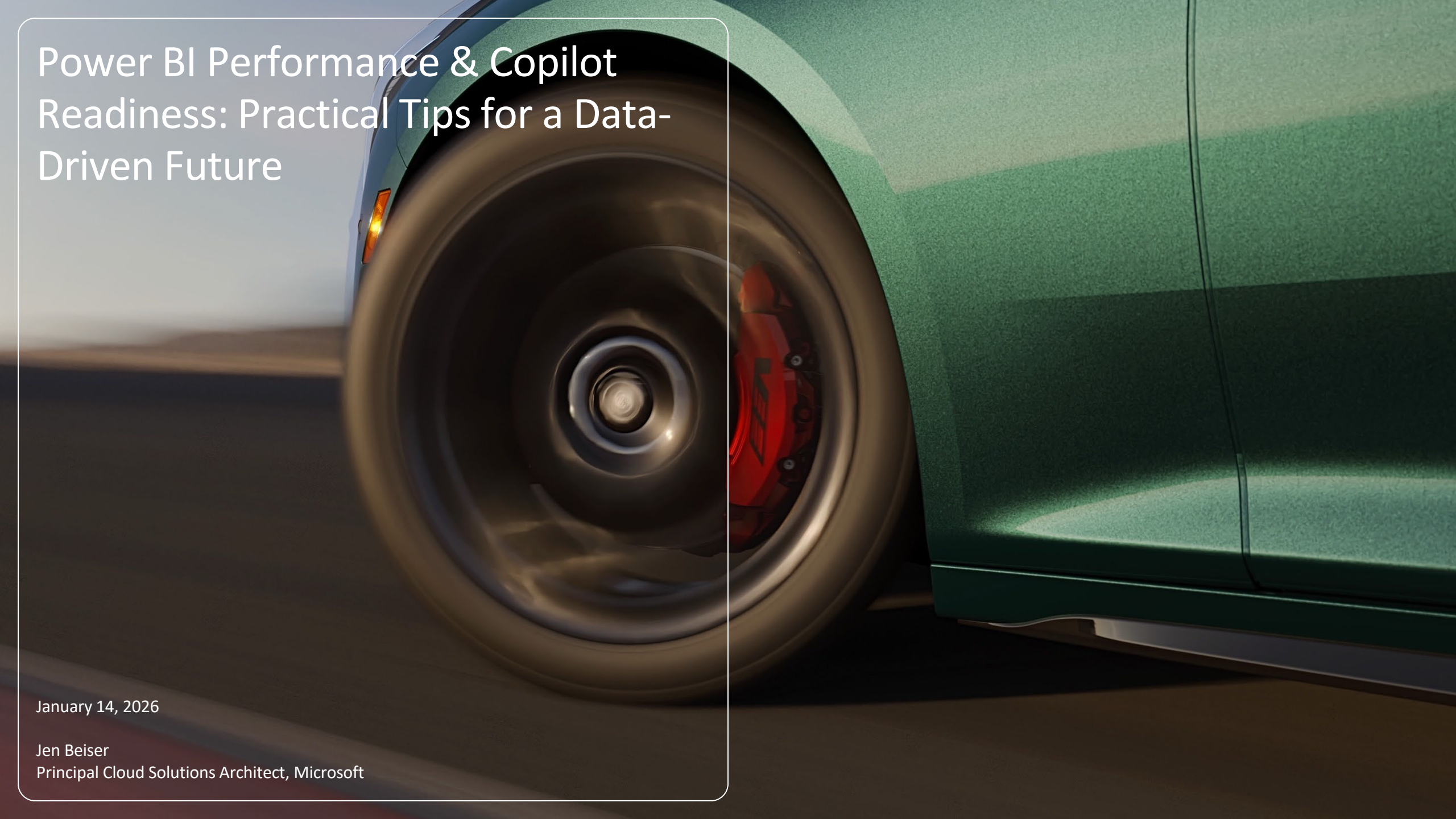




Power BI Performance & Copilot Readiness: Practical Tips for a Data- Driven Future

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Objectives

- Learn practical strategies to **optimize Power BI DirectQuery performance**
- Discover how to **design semantic models that enable Copilot** and AI-driven insights
- Engage through **demos and discussion**
- Leave with **actionable takeaways**



Agenda

1

Welcome and Context

Who's presenting and why this topic matters.

2

Direct Query Performance Tuning

See practical, high-impact tips that resonate with both business and tech users.

3

Semantic Model Design for Copilot

See how good modeling unlocks AI-driven insights.

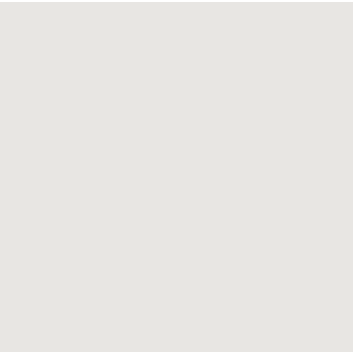
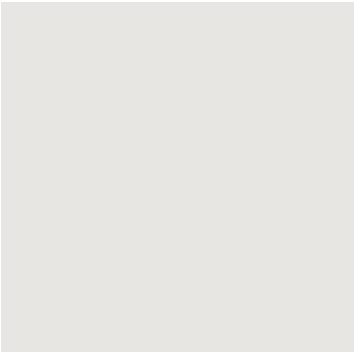


Speaker



Jen Beiser

Principal Cloud Solutions Architect





Today BI extends to everyone



Turning data into business insights is challenging

Common BI challenges include...

	End-to-end view	Data often resides in disparate locations, making it difficult to see a complete picture of your business
	Multiple data sources	Data residing in cloud solutions and on-premise locations is difficult to access and refresh securely
	Right data for the right users at the right time	Different roles have different needs and business users need the latest operational data

Gartner's 2025 Magic Quadrant

Gartner recognizes Microsoft as a leader in Analytics and BI platforms for 15 consecutive years. Why have the customers chosen Power BI?

- High Value
- Comprehensive analytics solution
- Centralized management
- Global scale
- Security, governance, and compliance

Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms

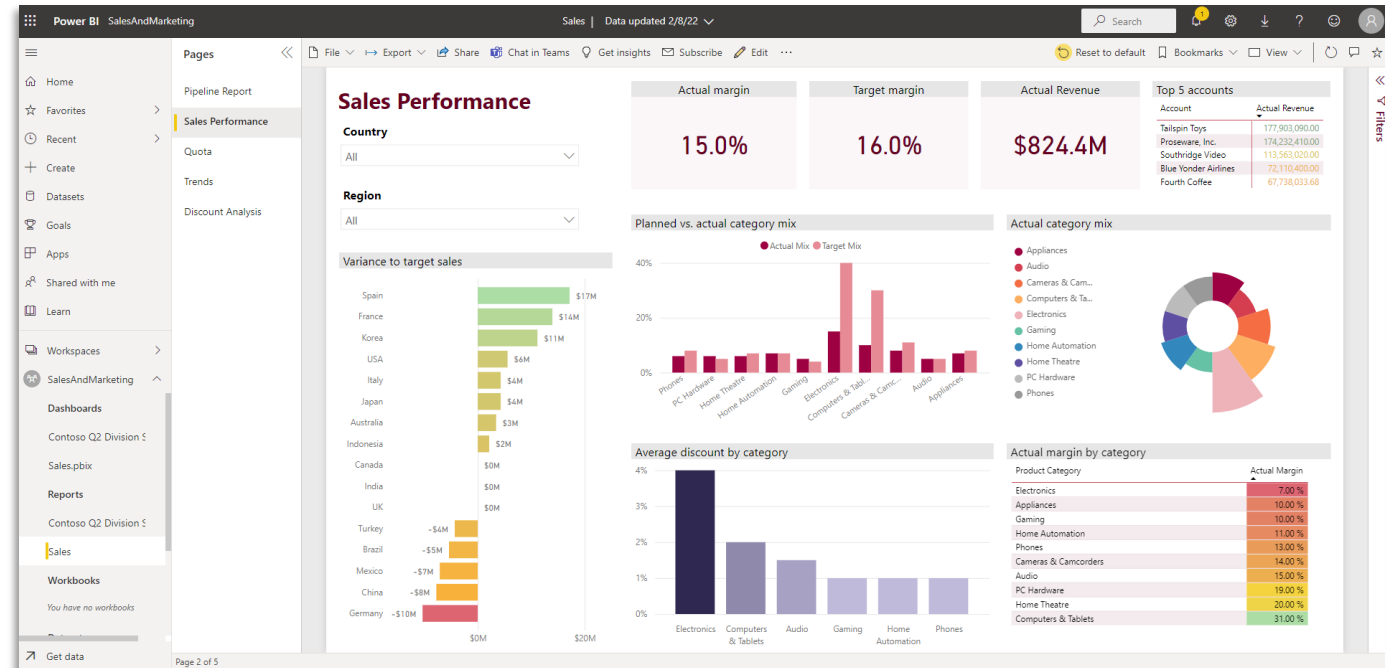


Power BI

Experience your data. Any data, any way, anywhere

Power BI is a cloud-based business analytics service that enables:

- Fast, easy access to your data
- Data discovery, exploration from any device
- Collaboration
- AI tools and Insights





For business users

- Ask questions about report content
- Summarize key insights in seconds

For tech users

- Write DAX queries
- Streamline semantic model documentation
- Provide a summary about a semantic model
- Suggest and create visualizations and report pages

Why Power BI performance and Copilot readiness matter?

They are **strategic enablers**.

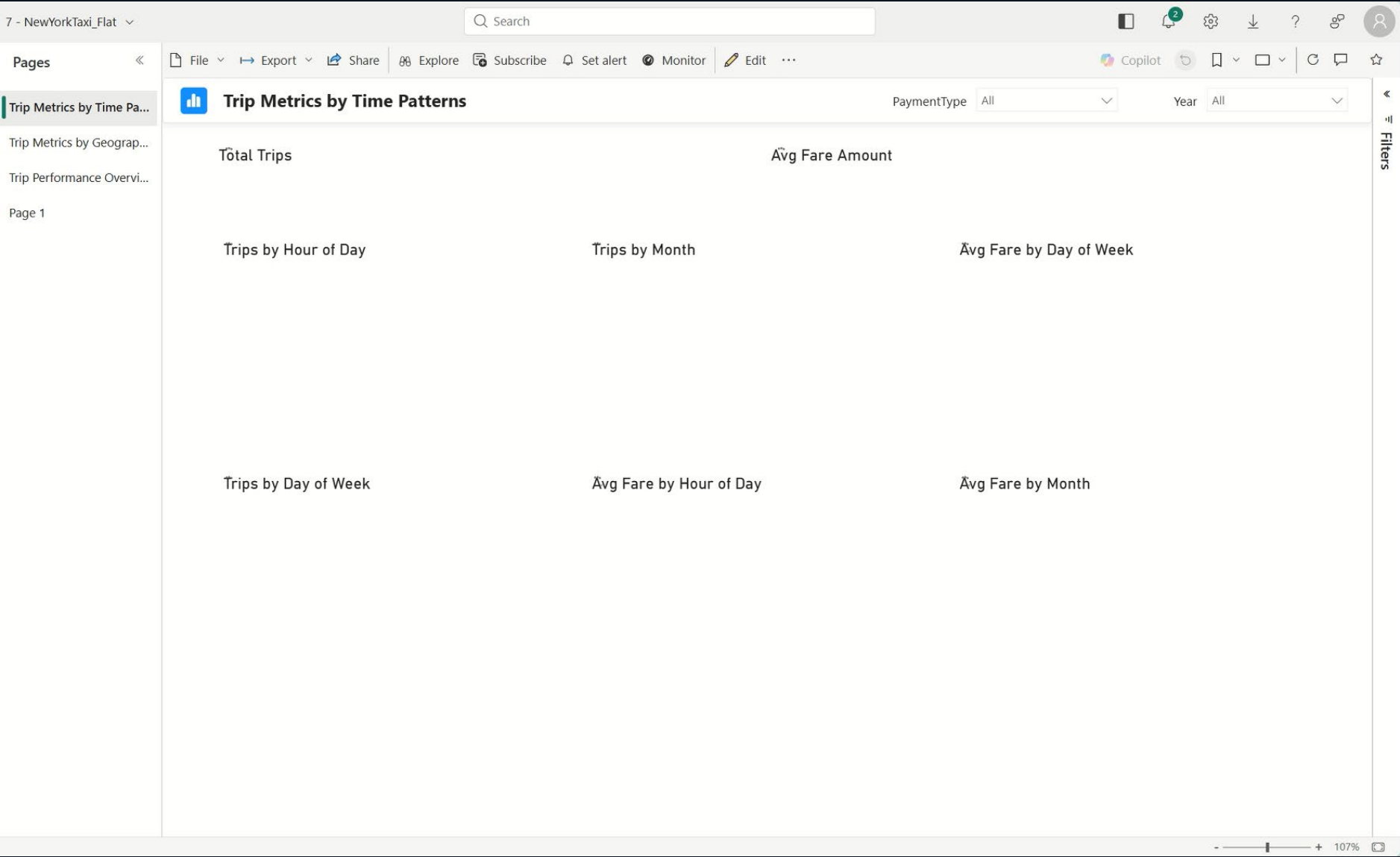
When PBI is optimized and AI-ready, the analytical data ecosystem becomes:

- Faster
- Cheaper
- Governed
- Accessible and
- AI enabled



DirectQuery Performance Tuning

users want to get the data they want when they want it



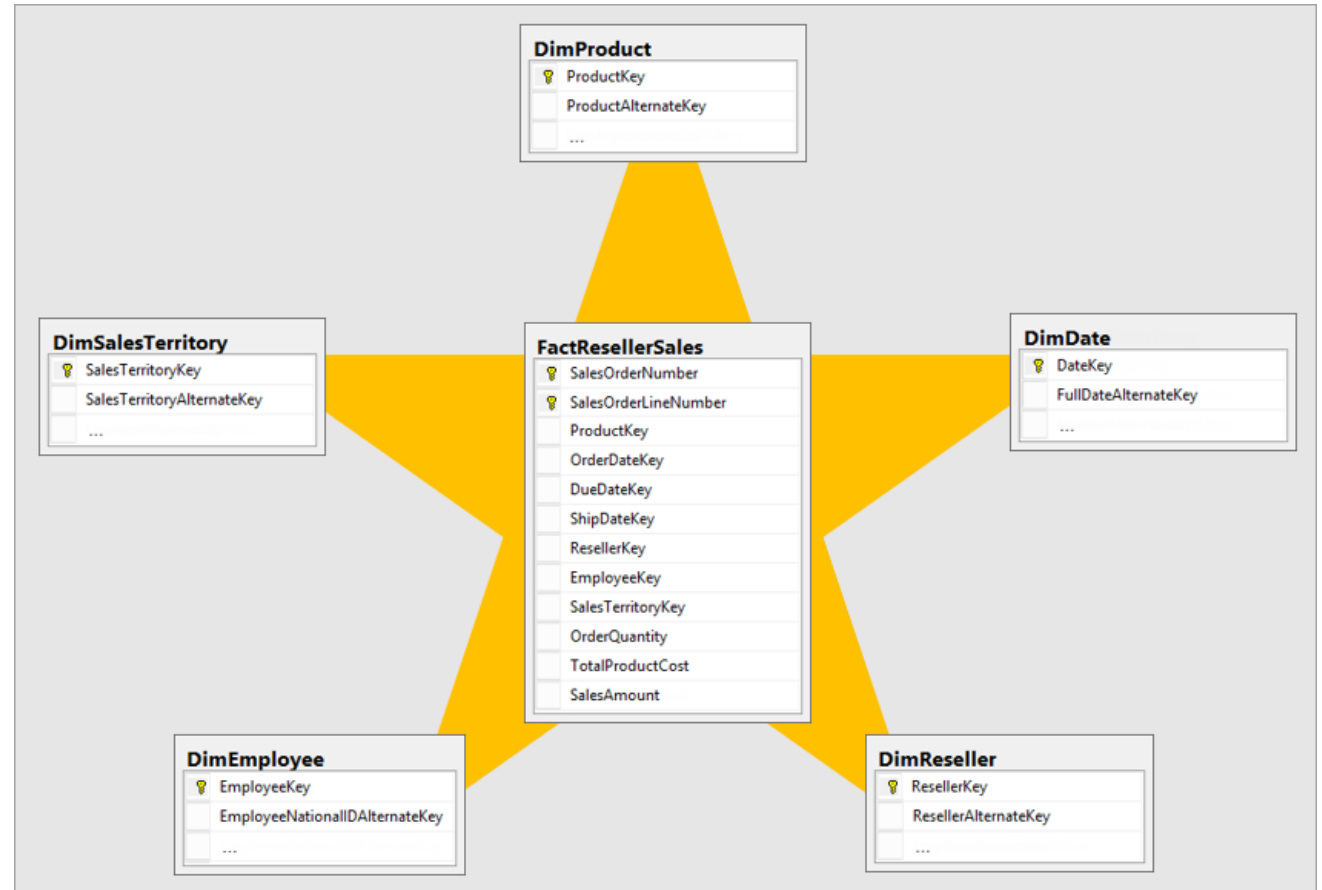


Tips to Optimize Performance

- ★ Star Schema
- ⚙️ Query Folding
- 📊 Referential Integrity and Data Reduction
- + User & Auto Aggregations
- 💻 Gateway and Capacity Settings
- 🔧 Tools

Performance Optimization ★ Star Schema

- Well-established data modeling approach
- Normalization vs Denormalization
- Dimension tables for business entities
- Fact tables for observations or events
- Simplified querying, enhanced performance, actionable insights



Performance Optimization Query Folding

What is
Query
Folding?



Query folding is the ability for a PQ query to generate a single query statement to retrieve and transform source data.



Query folding may occur for an entire query, or for a subset of PQ steps



Sources that support folding



Most data sources that have the concept of a query language support query folding. These data sources can include relational databases, OData feeds (including SharePoint lists), Exchange, and Active Directory



Native Query doesn't fold by default (more on this later)

Performance Optimization Query Folding

Can fold

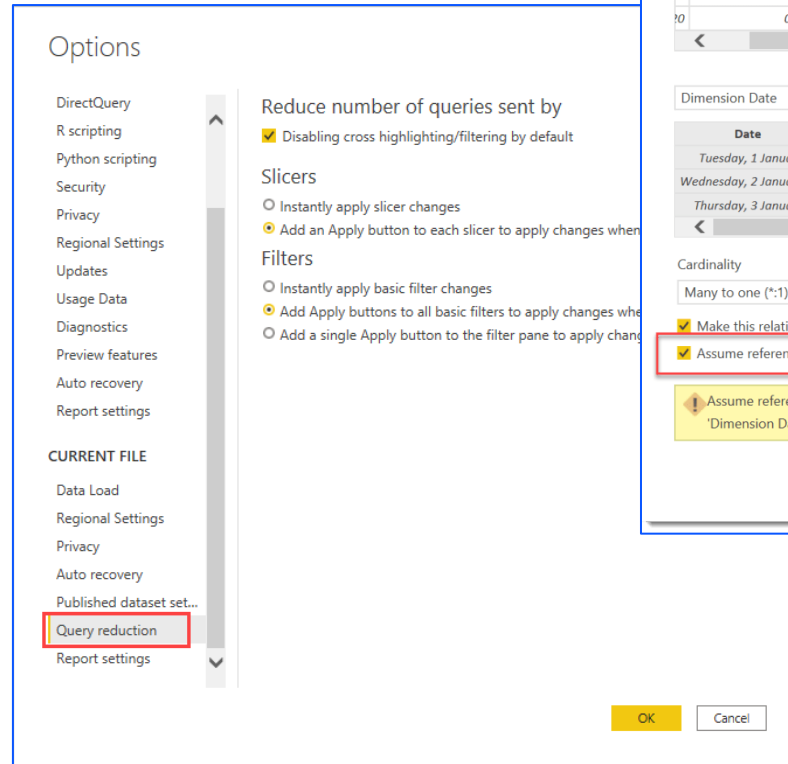
- Removing columns
- Renaming columns
- Filtering rows, with static values or Power Query parameters
- Grouping and summarizing
- Expanding record columns to achieve a join of two source tables
- Non-fuzzy merging of fold-able queries based on the same source
- Appending fold-able queries based on the same source
- Adding custom columns with simple logic
- Pivoting and unpivoting

Cannot fold

- Merging queries based on different sources.
- Appending (union-ing) queries based on different sources.
- Adding custom columns with complex logic
- Adding index columns.
- Changing a column data type.

Performance Optimization RI and Data Reduction

- Assume referential integrity (RI)
- Disable cross-highlighting and cross-filtering
- Apply button for slicers and filters
- Measure Killer for cleanup
- Vertipaq analyzer



Options

DirectQuery
R scripting
Python scripting
Security
Privacy
Regional Settings
Updates
Usage Data
Diagnostics
Preview features
Auto recovery
Report settings

CURRENT FILE

Data Load
Regional Settings
Privacy
Auto recovery
Published dataset set...
Query reduction
Report settings

Reduce number of queries sent by

- ☒ Disabling cross highlighting/filtering by default

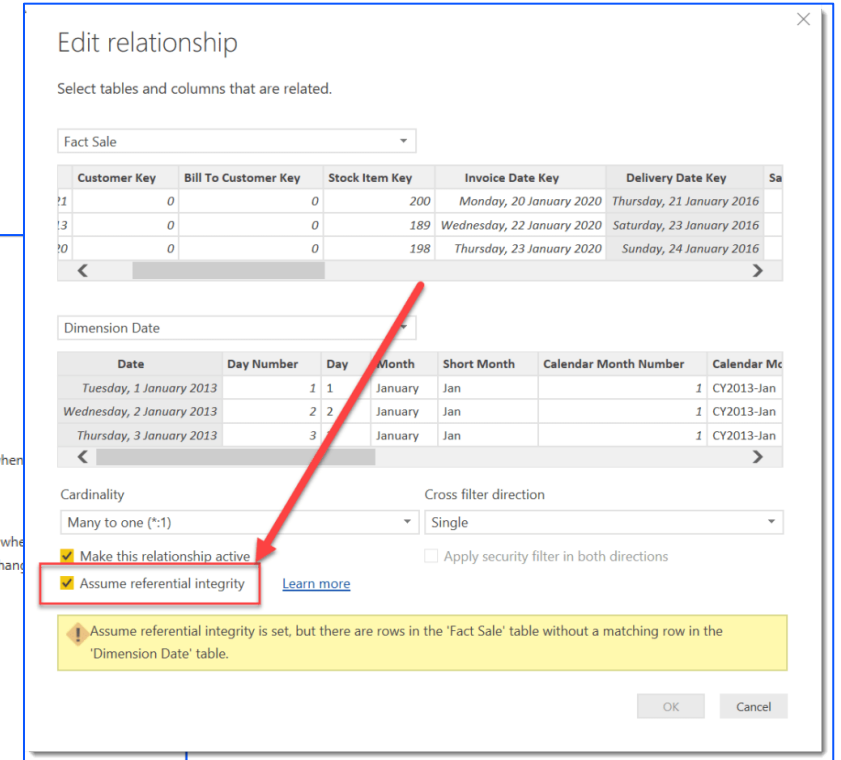
Slicers

- ☐ Instantly apply slicer changes
- ☒ Add an Apply button to each slicer to apply changes when

Filters

- ☐ Instantly apply basic filter changes
- ☒ Add Apply buttons to all basic filters to apply changes when
- ☐ Add a single Apply button to the filter pane to apply changes

OK Cancel



Edit relationship

Select tables and columns that are related.

Fact Sale

Customer Key	Bill To Customer Key	Stock Item Key	Invoice Date Key	Delivery Date Key	Sa
1	0	0	200	Monday, 20 January 2020	Thursday, 21 January 2016
3	0	0	189	Wednesday, 22 January 2020	Saturday, 23 January 2016
0	0	0	198	Thursday, 23 January 2020	Sunday, 24 January 2016

Dimension Date

Date	Day Number	Day	Month	Short Month	Calendar Month Number	Calendar Mc
Tuesday, 1 January 2013	1	1	January	Jan	1	CY2013-Jan
Wednesday, 2 January 2013	2	2	January	Jan	1	CY2013-Jan
Thursday, 3 January 2013	3	3	January	Jan	1	CY2013-Jan

Cardinality: Many to one (*:1)
Cross filter direction: Single

- ☒ Make this relationship active
- ☒ Assume referential integrity [Learn more](#)

Assume referential integrity is set, but there are rows in the 'Fact Sale' table without a matching row in the 'Dimension Date' table.

OK Cancel

Performance Optimization + User & Auto Aggregations

User Aggregations

- Cache aggregated data in-memory
- Reduce computational cost for direct query
- Configure based on query patterns
- Import, Direct Query, and Dual Mode
- Fine-grained control

Manage aggregations

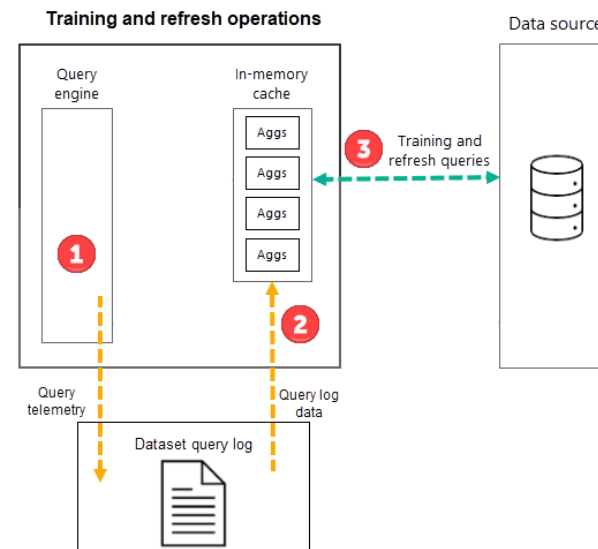
Aggregations accelerate query performance to unlock big-data sets. [Learn more](#)

Aggregation table: Sales Agg Precedence: 0

AGGREGATION COLUMN	SUMMARIZATION	DETAIL TABLE	DETAIL COLUMN
OrderDateKey	GroupBy	Sales	OrderDateKey
CustomerKey	GroupBy	Sales	CustomerKey
ProductSubcategoryKey	GroupBy	Product	ProductSubcategory...
SalesAmount_Sum	Sum	Sales	SalesAmount
UnitPrice_Sum	Sum	Sales	UnitPrice

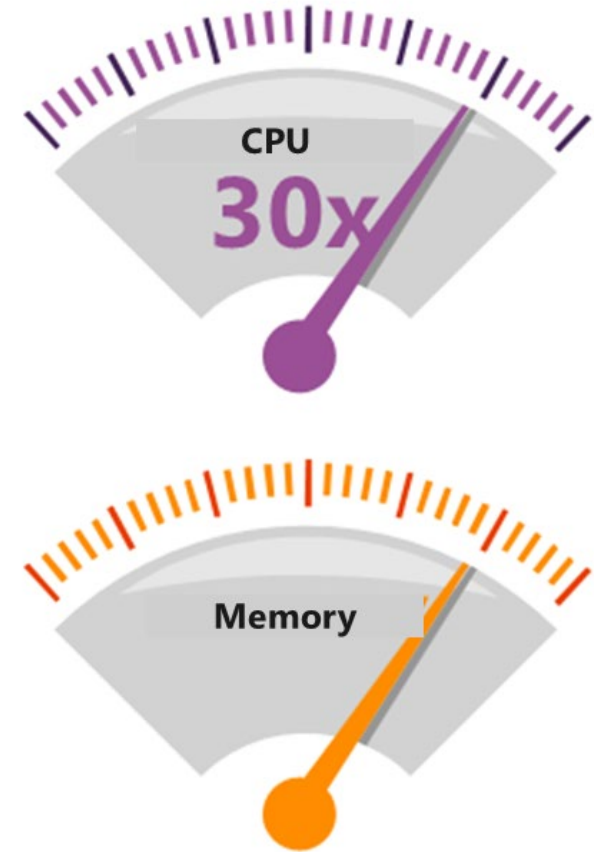
Auto Aggregations

- Cache aggregated data in-memory
- Reduce computational cost for direct query
- Built on the manual infrastructure
- Created by AI
- Easy set-up
- Fine tuning



Performance Optimization Gateway and Capacity Settings

- **Update** to the latest version (only last 6 versions support
- **Separate import and direct query**
- Deploy **multiple nodes with load balancing**
- Ensure **adequate CPU sizing** (start with 8 cores)
- Configure Spooler Directory and Paging File to use **SSD drives**
- **Check network latency** with ping (gateway to data source and Azure Service Bus)
- Avoid installing On-Premises Data Gateway and ADF Self-Hosted Runtime on the same machine
- [Gateway Performance PBI template](#)





Performance Optimization **Tools**

- Performance Analyzer
- DAX Studio
- Vertipaq Analyzer
- Measure Killer
- Diagnostics (Import)



Demo.



Copilot in Power BI

Tips to ensure users have the best experience when they use Copilot



What is Copilot in Power BI?

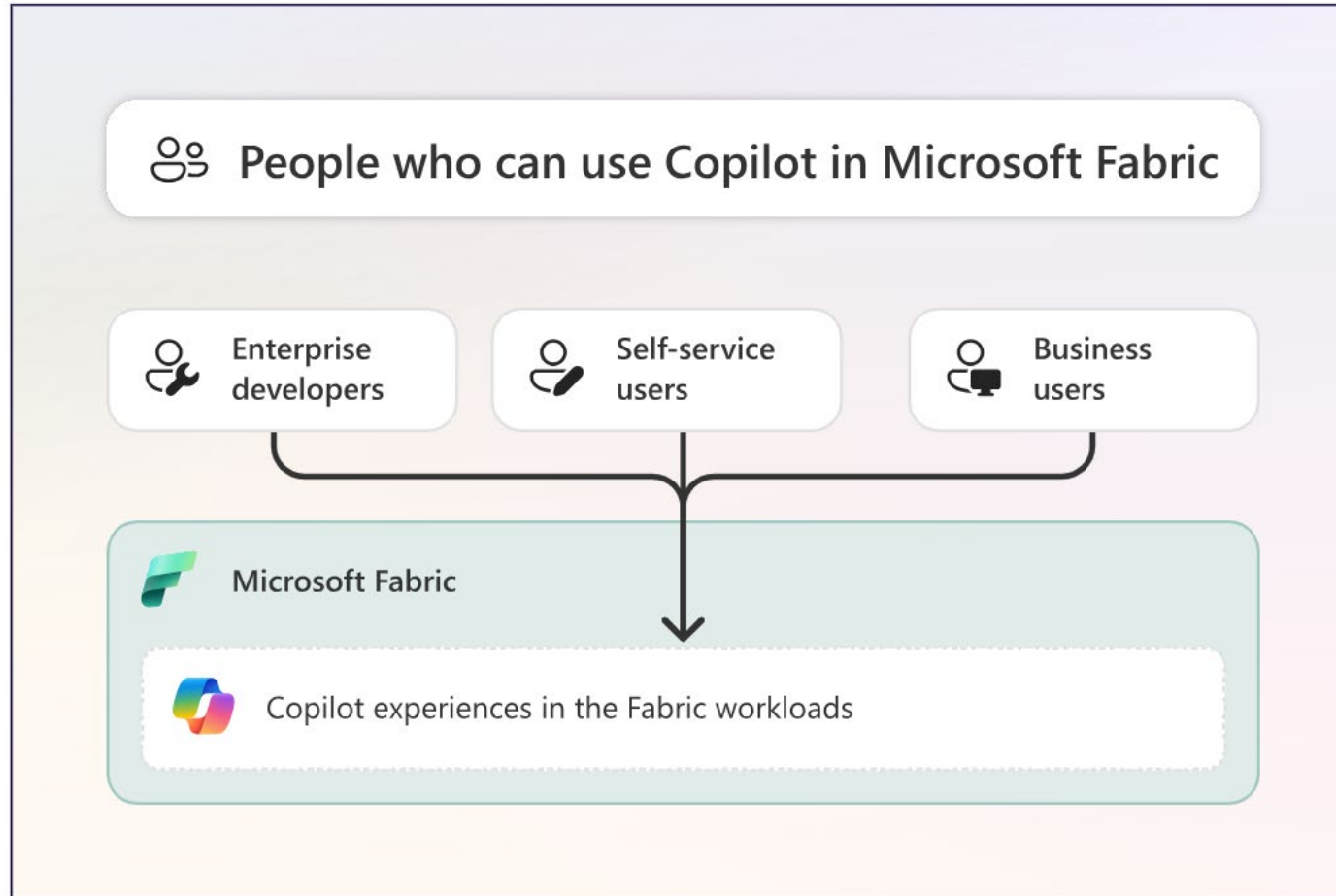


Generative AI assistive technology that uses large-language models (LLMs) that facilitate user interaction with their Power BI Reports

Important

Copilot in Fabric aims to *augment* the abilities and intelligence of human users. Copilot can't, and doesn't, aim to *replace* the people who today create and manage reports or other Fabric items. To get the most out of Copilot in Fabric, you should consider how you'll enable these individuals and teams to make their work more efficient or improve their outcomes.

Who can use Copilot in Power BI?





Semantic Model Design for Copilot

Good modeling unlocks AI-driven insights



Tips to Optimize for Copilot



Clear Names



No Abbreviations



Relationships & Hierarchies



Default Categories and Aggregations



Gateway and Capacity Settings



Mindful Data Types & Formatting



Time Intelligence



Copilot Design Tip 🚫 Clear Names

Name fact tables based on actions or events

Copilot interprets table names semantically. Action-oriented names help it understand the purpose of the data, improving the relevance of generated insights.

Copilot Design Tip **No Abbreviations**

Rename abbreviated columns

Full names improve clarity and help Copilot understand the semantic meaning of each field.

Copilot Design Relationships & Hierarchies

Ensure all fact tables have relationships

Relationships allow Copilot to traverse the model intelligently. Without them, it may misinterpret data or fail to generate meaningful summaries.

Copilot Design Tip **Default Categories and Aggregations**

Set default categories for location data

Helps Copilot recognize location fields for mapping and geo-based analysis.

Review default aggregations

Prevents misleading insights and ensures Copilot applies the correct logic.

Copilot Design Tip **Default Categories and Aggregations**

Hide key columns

These are useful for relationships but not for analysis. Hiding them de-clutters the model and guides Copilot toward meaningful fields. This is easily done through Prep for AI.

Use measure killer to clean up

Reduces model complexity and size, making it easier and more efficient for Copilot to navigate and generate accurate results.



Copilot Design Tip **Mindful Data Types & Formatting**

Validate data types and formatting

Proper data types help Copilot interpret fields correctly and apply appropriate operations. It also has performance benefits when going from text to numeric if numeric data is stored as text.

Copilot Design Tip Time Intelligence

Include Time Intelligence Measures

These measures enable Copilot to analyze trends and identify key influencers over time. Without them, Copilot may only describe changes without explaining drivers.

For example, in a model without time intelligence, Copilot might say:

"The decline in variable profit for Product X from Q1 to Q2 2025 is evident, with average profit dropping from \$X in January to \$X in April, \$X in May, and \$X in June. The data does not provide explicit reasons for this decline."

Including time-based metrics gives Copilot the context it needs to surface meaningful insights."



Optimized for Copilot

Demo.

Thank you!